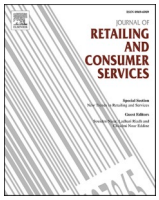
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Artificial intelligence in live streaming: How can virtual streamers bring more sales?

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ABSTRACT

Virtual streamers are gaining traction in the thriving live streaming commerce with the support of artificial intelligence technology. While virtual streamers have excellent interactive capabilities, effectively leveraging them to enhance sales remains a significant challenge for companies. This research analyzes 1,960,444 real live streaming comments from 30 brands and conducts three experimental studies and half-structured interviews for this dilemma. The findings reveal that consumers exhibit a higher tendency to seek promotional information (vs. product information) when engaging with virtual streamers (vs. human streamers). This unique shopping mentality is mediated by consumers' motivation inference about the company. Consequently, aligning virtual streamers with promotional products and human streamers with new products can be beneficial in driving sales by catering to diverse consumer information preferences. These findings provide valuable insights into effectively adopting virtual streamers, optimizing recommendation strategies, and understanding consumers' subjective perceptions.

1. Introduction

In today's world where a profound digital transformation driven by Artificial Intelligence (AI) is taking place (Mostafa and Kasamani, 2022), companies are deploying AI-generated virtual streamers into the thriving realm of live streaming commerce (LSC) to boost operational efficiency and enhance the customer experience (Chen et al., 2024; Shao, 2024; Gao et al., 2023; Hu and Ma, 2023). Virtual streamers represent digital entities with anthropomorphic features, generated through computer algorithms and driven by artificial intelligence (Gao et al., 2023; Miao et al., 2022; see Fig. 1). They emulate human streamers by engaging in product recommendations, interacting with consumers in real time and providing shopping guides through algorithmically synthesized avatars (Yao et al., 2024; Gao et al., 2023; Hu and Ma, 2023). Industry statistics show that virtual streamers are experiencing rapid growth, China's virtual avatar industry was worth 254.161 billion yuan in 2023, and it is expected to reach 523.16 billion yuan in 2025 (Zhiyanzhan Research, 2024).

Companies considering the adoption of virtual streamers encounter the crucial quandaries: Are they sufficiently informed about the distinctive attributes of the new role of virtual streamers? Should the live streaming strategies for virtual streamers mirror those employed for human streamers? Currently, the prevailing approach among companies involves substituting virtual streamers for human streamers, resulting in both roles presenting the same products and content. We argue that this is inappropriate, as consumers perceive human and AI characters differently, leading to diverse psychological responses that influence interaction intentions and behavioral outcomes (Wan and Jiang, 2023). While the integration of AI technology in customer service has shown promise in reducing costs (Sands et al., 2022; Ling et al., 2021; Tsai et al., 2021; Williams, 2019) and improving communication efficiency (Puntoni et al., 2021; Syam and Sharma, 2018; Shao, 2024), instances of consumer disappointment and negative psychological reactions towards AI and virtual streamers are prevalent (Sands et al., 2022; Stein and Ohler, 2017). These divergent effects pose challenges to companies in making effective strategies for deploying virtual streamers.

Against this backdrop, our research focuses on how virtual and human streamers will affect consumers' shopping psychology and contribute to the company's sales growth. Our findings indicate that consumers interacting with virtual streamers exhibit a higher tendency to seek promotional information, reflecting the "saving money" mentality of consumers. This tendency stems from consumers inferring cost

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Fig. 1. Virtual streamers.

reduction motivations about companies, leading to perceptions of price unfairness and a quest for monetary compensation in the live streaming room. Conversely, the company's motivation in adopting human streamers will be perceived as improving service, causing consumers to prioritize product-related information such as features and quality. Consequently, virtual streamers are more suitable for recommending promotional products, whereas human streamers are better suited for selling products requiring detailed information dissemination to consumers (e.g., new products). To support our point, we use five studies, including a text analysis of 1,960,444 real consumer live comments from 30 branded live streaming rooms, showcasing the critical importance of a rational allocation of product recommendation strategies between virtual and human streamers to effectively capitalize on the marketing opportunities presented by LSC.

By exploring the utilization of virtual streamers in novel LSC marketing scenarios, this research makes several major contributions. We extend research on phycology of human-technology interaction (Puntoni et al., 2021; Davenport et al., 2020). We find that interacting with virtual streamers makes consumers more interested in promotional information, and virtual streamers are more suitable for recommending promotional products to consumers. These findings deepen the understanding of how consumers conditionally embrace virtual digital representations. Another theoretical contribution related to LSC (Gao et al., 2023; Lu and Chen, 2021). The existing literature provided scant description and discussion of virtual streamers, while we introduce innovative investigations into tailored product recommendation strategies for virtual streamers.

2. Literature review

Research on consumer-streamer interactions has been accelerating as LSC has become a powerful marketing tool for organizations (e.g. Tian and Frank, 2024; Luo et al., 2024; Li et al., 2021). In typical LSC, a streamer establishes a virtual space - the live-streaming room - to showcase products, allowing consumers to freely join and interact with the streamer through live comments (Zhang et al., 2024). Extant research on LSC consists of two main aspects. Those exploring the impact of consumer's own characteristics and psychological factors on the shopping process from a consumer perspective (see Web Appendix A; Table W1), and those elucidating how human streamer characteristics influence consumer purchasing behaviors from the streamer's standpoint (see Web Appendix A; Table W2). Based on the review of existing literature, it is evident that research on live streaming strategy of virtual streamers was notably scarce. Additionally, there was a notable absence of comparative studies exploring the influence of virtual streamers versus human streamers on consumers' psychological responses. Therefore, it is necessary to conduct in-depth discussion on virtual streamer.

The novelty of virtual streamers and their potential impact on consumer experiences drove this research. Virtual streamers offer distinct advantages over human streamers in enhancing the consumer shopping experience, notably evident in cost-effectiveness (Thomas and Fowler, 2021), public opinion safety (elimination of subjective errors and potential scandals, Gao et al., 2023), swift responsiveness and heightened service efficiency (Sun and Tang, 2024; Gao et al., 2023), and friendliness (devoid of negative emotions or work fatigue, Wu et al., 2023). However, virtual streamers also face the same ethical issues inherent in AI roles, such as AI bias (Du and Xie, 2021), consumer privacy (Wirtz et al., 2018), and unemployment (Khogali and Mekid, 2023). Table 1 lists the limited research that has been conducted on virtual streamers. Notably, these few literatures focused on examining the impact of virtual streamers' language, emotion, and appearance characteristics on consumers' purchase and usage intentions (e.g. Yan et al., 2024; Xie et al., 2024; Chen et al., 2024). Thus, we further find that there are two shortcomings in the research on virtual streamers. The first is the lack of attention to the fundamental difference between virtual and human streamers - the difference in identity - on the impact of consumers. Second, there is an obvious lack of research on the product recommendation strategies of virtual vs. human streamers. Based on this, we would like to enhance the comparison between virtual and human streamers and show how they lead to diverse psychological responses from consumers.

Taken together, given the burgeoning influence of AI agents akin to human agents (Wan and Jiang, 2023), enhancing comprehension of the distinct attributes of virtual streamers and elucidating consumer responses are crucial for fostering the efficient advancement of LSC. We will specifically discuss product presentation strategies for different streamer types and the driving mechanisms in the next sections.

3. Theoretical background and hypothesis development

3.1. Attribution theory

To better understand consumers' responses when interacting with virtual or human streamers, we begin by exploring consumer psychology from a theoretical perspective. Attribution theory aims to clarify how people make causal explanations of events and how these explanations influence behavioral outcomes (Kelley, 1971). Making causal attributions is an inherent need for individuals to make sense of social events (Kelley, 1971). Consumers' attributions can emerge spontaneously on the basis of their observable cues (Kramer et al., 2012), as the consequence of their perceptions of a firm's effort to serve its customers (Morales, 2005). Attribution theory suggests that when an event is unexpected or the outcome is negative, people look for the cause of the event (Weiner, 1985).

3.2. Expectation-confirmation theory

Expectation-confirmation theory posits that consumers assess a service based on their initial expectations and the confirmation of those expectations through actual performance (Oliver, 1980). Consumers use their expectations as a benchmark for evaluating a service, and their perception of service performance is shaped after using the service. Expectation confirmation occurs when service performance matches expectations (Oliver, 1980). If the service outcome surpasses (falls short of) expectations, consumers develop a more positive (negative) perception (Oliver, 1980). Aligning with this theory, both confirmation of expectations and superior service performance lead to positive customer perceptions (Anderson and Sullivan, 1993).

3.3. Consumer information-seeking behavior in LSC

AI integrated into live streaming environments represents a significant innovation that introduces consumers to novel stimuli (Gao et al.,

Table 1
Summary of virtual streamer research.

Source	Characteristics of Virtual Streamers	Consumer's Psychological Reaction	Method and Participants	Downstream Consequences
Yan et al. (2024)	Language	Consumers perceive virtual streamers as less capable of conveying sensory information	Experiment, consumers	Virtual streamers are less effective than human streamers in recommending hedonic items
Yao et al. (2024)	Language	The specific language style of the streamer will trigger consumer stereotypes	Experiment and qualitative research, consumers	When virtual streamers use social-oriented language, consumer's willingness to purchase hedonic products will increase
Hu and Ma (2023)	Language	When virtual streamers use sensory language, it will cause consumers' language expectation violation	Experiment, consumers	The use of sensory language by virtual streamers will lead to a decrease in consumers' purchase intention
Xie et al. (2024)	Emotion	virtual streamers suppress emotional fusion	Experiment, consumers	Consumer brand forgiveness will higher when virtual streamers make inappropriate remarks
Chen et al. (2024)	Appearance	Anthropomorphic of virtual streamers affect consumer's psychological distance and higher trust	Experiment, consumers	Anthropomorphic of virtual streamers improve consumers' willingness to accept
Sun and Tang (2024)	Appearance	When virtual streamers have low form realism, consumers anticipate its behavioral capabilities to be scripted and mechanized	Experiment, consumers	The behavioral realism of virtual streamers positively impacts consumer purchase intentions when the form realism of virtual streamers is low
Ji (2024)	Appearance, expertise	No detailed analysis	Secondary data, consumer	The trustworthiness, attractiveness, and expertise of virtual streamers have a positive impact on sales performance
Gao et al. (2023)	Appearance	The likeability, animacy, and responsiveness of virtual streamer increase social presence and telepresence perceived by consumers	Survey, consumers	The likeability, animacy, and responsiveness of virtual streamer have a positive impact on consumers' purchase intention
Shao (2024)	Effectiveness and ease of use	No detailed analysis	Survey, consumers	Innovation resistances influence consumer's switching intentions to virtual streamers
Xu et al. (2024)	Personalization	No detailed analysis	Survey, consumers	The personalization of virtual streamer will impact brand image
Wang et al. (2023)	Social presence	Perceived social presence will affect consumers' trust	Experiment, consumers	Social presence influences trust for live streamers, but it only directly affects purchase intention towards virtual streamers rather than human streamers
Wu et al. (2023)	Sociability	high-social virtual live streamers enable consumers to perceive a higher social presence	Experiment, consumers	The sociability of virtual streamers improves the experience value of consumers
Wan and Jiang (2023)	Effectiveness and usefulness	Consumers perceive lower warmth, trust, usefulness, and dialogue of virtual streamers	Experiment and qualitative research, consumers	Consumers have fewer positive attitudes toward virtual streamers, and virtual streamers are less helpful in consumer purchasing decisions
This research	Identity	When consumers are exposed to virtual streamers, they will be more inclined to "save money"	Secondary data, experiment and qualitative research, consumers	Virtual streamers recommending promotional products will increase consumers' purchase intention

2023; Wan and Jiang, 2023), prompting motivated inferences about the company. Virtual streamers, as a type of AI agents, are perceived as disruptive or groundbreaking innovations (Tsai et al., 2021; Belanche et al., 2020). We argue that such innovations capture the attention of consumers. Besides, consumers participate in value co-creation in live streaming and become co-producers of services (Zhang and Xu, 2024). And the advent of virtual streamers has dramatically reshaped service engagement (Chen et al., 2024; Shao, 2024; Wu et al., 2023), making consumers notice the significant changes in interaction objects. Consequently, consumer tend to be very sensitive to the changes in service exchanges that result from virtual streamer introductions. According to attribution theory (Weiner, 1985), unexpected event makes individuals infer underlying reasons, a phenomenon often reflected in consumer live comments. For instance, comments like "It is amazing that AI can conduct live streaming" or "Why utilize robots for live streaming?" demonstrate consumer curiosity and perplexity regarding the rise of virtual streamers, with consumers linking this trend to company motives.

Companies often navigate a trade-off between enterprise performance metrics (e.g., service productivity) and customer-oriented metrics (e.g., service quality) when implementing automation strategies (Rust and Huang, 2012). In practical settings, the reasons behind companies' adoption of virtual streamers vary based on their unique business requirements - driven by cost reduction, service improvement, or a combination of both (Castelo et al., 2023). However, consumer attributions about a firm's adoption of virtual streamers are not always consistent with the firm's original intentions. The economic benefits of AI technology have led to the widespread holding of lay beliefs (commonsense explanations of the external world; Molden and Dweck, 2006) about low cost. It is intuitively assumed that the adoption of virtual streamers aims at cost reduction and time savings (Yao et al., 2024; Gao et al., 2023), given that AI technology significantly reduces organizational costs (Ling et al., 2021; De Keyser et al., 2019). Additionally, consumers' profit-seeking stereotypes about for-profit firms happen to coincide with the above beliefs.

Existing research indicates that profit-seeking firms are perceived as prioritizing profits over consumer well-being, shaping consumers' judgments about for-profit firms' intentions (Bhattacharjee et al., 2017). Negative intuitions about corporations are rooted in the stereotype that business organizations behave unethically in pursuit of profits (Freund et al., 2024). These views provide insights into how consumers perceive firms' motives based on initial impressions - profit-seeking, rather than focused on benefiting consumers. Thus, consumers perceive companies' deployment of virtual streamers as motivated by profit considerations, especially considering the evident cost-saving benefits associated with virtual streamer adoption. Specifically, we hypothesize that consumers tend to interpret companies' adoption of virtual streamers as driven by cost reduction motives compared to the utilization of human streamers.

Then, two streams of literature help to explain the shifts in consumers' shopping mindset following motivational inferences. Research on price/cost perception highlights that consumers instinctively factor in the costs and profits when evaluating products' price (Bolton et al., 2003) and often hold beliefs about the impact of labor costs on product pricing (Wongkitrungrueng et al., 2020). The reduction in perceived costs diminishing consumers' perceptions of price fairness (Bolton et al., 2003). In response, consumers seek monetary compensation to safeguard their financial interests, driven by a goal to protect themselves financially and enhance their own welfare (Xia et al., 2004). In the

context of LSC, seeking promotional information is a way for consumers to seek monetary compensation. For instance, it is common to see live comments such as "Is there a coupon for this?" or "Are there any promotions?" reflect consumer's tendency to seek monetary benefits.

Similar logic can be supported by the profit view of zero-sum game, where increased profits for one party reduce the value received by the other (Bhattacharjee et al., 2017), even though both parties may benefit from the transaction. Therefore, when consumers perceive that companies reduce labor costs through the adoption of virtual streamers, they infer that the company's profits increase, potentially at the expense of consumer welfare. In seeking to protect their interests, we propose that consumers tend to seek promotional information when engaging with virtual streamers.

In contrast, human streamers prompt consumers infer the company intends to improve service and focus on product information. Given that product uncertainty has been identified as a key barrier to consumer purchasing online (Kim and Krishnan, 2015), human streamers reduce product uncertainty and online information asymmetry by creating dynamic presentations and providing rich and vivid product information (Chen et al., 2023; Gu et al., 2023), which led consumers to associate human streamers with the company's efforts to improve service (Zhou and Huang, 2023; Lu and Chen, 2021). Human streamers enhance service and demonstrate products for consumers in an immersive and intuitive manner, which raises consumers' attention to the effect, quality, and suitability of products (Chen et al., 2022). Therefore, we argue that consumers heightened attention towards product information when interacting with human streamers. Based on the above, we propose the following hypotheses:

H1. Consumers exposed to virtual streamers (vs. human streamers) will show a higher tendency to seek promotional information (vs. product information).

H2. Consumers infer that companies adopting virtual streamers (vs. human streamers) are primarily motivated by cost reduction (vs. service improvement), influencing information-seeking behavior as delineated in H1.

3.4. The role of product category

Based on consumers pay more attention on promotional information (vs. product information) when interact with virtual streamers (vs. human streamers), the next question revolves around how streamers can drive sales growth. According to the expectation-confirmation theory (Oliver, 1980), consumers have expectations for the information provided by the streamers, and if streamers can meet the expectations, it will help to improve consumer's purchase intention. Specifically, consumers show expectations for promotional information (vs. product information) when facing virtual streamers (vs. human streamers). Therefore, we anticipate that live streaming will yield more pronounced sales growth when streamers actively provide consumers with information that they are expect.

We illustrate this through two product categories. Firstly, we examine whether virtual streamers recommending promotional products can achieve a more substantial sales uplift than human streamers. Promotional products refer to products that are using monetary or non-monetary sales promotion tools to encourage consumers' purchase (Sinha and Verma, 2020). Here, we do not deny the effectiveness of human streamers recommending promotional products. It is essential to note that promotional products typically enhance consumer's purchase intention in generic scenarios (Sinha and Verma, 2020; Carlson and Kukar-Kinney, 2018). However, consumers will activate the "money--saving" mentality when they encounter with virtual streamers. According to expectation-confirmation theory, meeting consumers' expectation of "saving money", that is, virtual streamers recommending promotional products, will significantly increase consumers' purchase intentions.

Secondly, we provide further evidence by testing the importance of human streamers in recommending new products. A new product is one that has qualities or attributes that are novel or unfamiliar to consumers (Faraji-Rad et al., 2017), which makes consumers face a high degree of perceived uncertainty and requires more details (Castaño et al., 2008; Hoeffler, 2003). We propose that, in the case where consumers are more interested in human streamers presenting product information, according to the expectation-confirmation theory (Oliver, 1980), delivering rich product information to consumers by human streamers helps reduce consumers' perceived uncertainty and meet consumers' expectations for product information when a company launches a new product to the market. Such a recommendation strategy is more helpful to improve consumers' purchase intention for new products. Taken together, we hypothesize the following:

H3. Virtual streamers (vs. human streamers) are more effective in increasing sales in terms of presenting promotional products.

H4. Human streamers (vs. virtual streamers) are more effective in increasing sales in terms of presenting new products.

4. Methodology

Our conceptual framework is shown in Fig. 2. Across five studies, combined with real consumer live comments data and experiments, we tested the hypotheses of this research. In Study 1, we analyze a data set of 30 brands for one month, focusing on live comments sent by consumers. We extract and cluster the topics of comments and find that consumers asked more promotional information in virtual streamer's room, while they were more interested in product information in human streamer's room (consistent with H1). In Study 2, the first of three experiments, we replicate the conclusions in Study 1 and confirm this effect is caused by consumers' motivational inferences when exposed to different types of streamers (consistent with H1 and H2). Subsequently, we examine what type of product is suitable for different types of streamers to recommend (Study 3 and Study 4). Study 3 finds that virtual streamer's recommendation of promotional products will significantly increase consumers' purchase intention (H3). Study 4 shows when companies promote new products, using human streamers to recommend to consumers will be more beneficial to increasing purchase intention (H4). All results in Study2~4 was analyzed using SPSS 29. Finally, in Study 5, we conducted half-structured interviews with 20 participants to improve the external validity of the research.

5. Study 1

Study 1 analyzed real data sets from the live streaming rooms of the e-commerce platform (Taobao Live) of 30 brands. Taobao Live is China's largest and most technologically advanced live streaming platform. The main purpose of this study is to provide preliminary evidence of consumer information-seeking behavior in live streaming rooms.

5.1. Data

The data set was provided by a professional live streaming monitoring company based on crawler technology. We sampled brands that hired both virtual and human streamers during the same period to build consumer live comments data set. The data set covered 4,061,987 consumers comments of 1537 live streaming rooms from 30 brands (industries include beauty care, clothing and footwear, sports and outdoor, home decoration, food, and digital appliances) from September 20 to October 20, 2021. Each brand did not use virtual and human streamers on a fixed daily basis. Sometimes only human streamers introduce products, and sometimes only virtual streamers were active during the whole day. Therefore, in our data set, there were a total of 788 live streaming by human streamers and 749 records by virtual streamers.

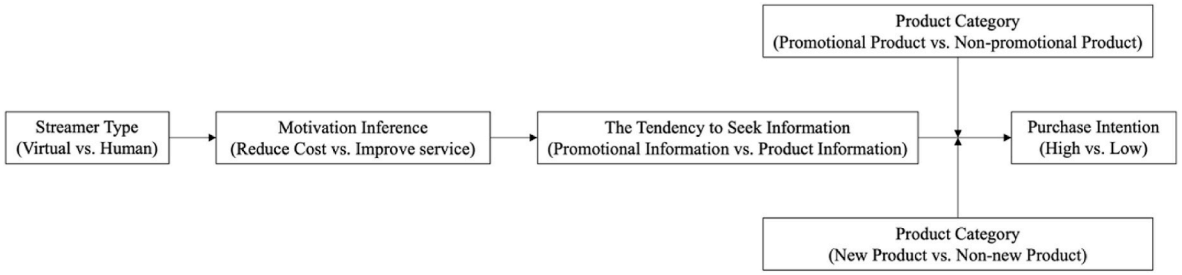


Fig. 2. Conceptual framework.

We focused on two topics in consumer's comments: promotional information and product information. We employed Word2Vec (Mikolov et al., 2013) to generate word embedding and then determine the degrees of similarity between two reviews. Next, the K-means algorithm was selected to cluster the keywords and categorize comments. Finally, the keywords were clustered into several sets we focused on.

Before analyzing the comments text, we performed the following four preprocessing procedures to remove non-informative content and interference information to improve computational reliability (Ji et al., 2023). 1) Convert all text content to Chinese and remove punctuation marks; 2) Meaningless numbers, emoticons, spaces, and stop words are removed; 3) Remove repeated comments generated continuously by the same user; 4) Remove comments automatically generated by the system. Finally, a total of 1,960,444 comments participated in the data analysis. All comments are presented in Chinese. Table 2 shows the data preprocessing process in detail.

We were particularly interested in comment content as it mirrors the extend of consumer focus on specific types of information. For instance, a higher proportion of promotional information in live comments within a live-streaming room suggests a higher tendency to seek promotional information of consumers, supporting our H1. As illustrated with examples from brands in the clothing and footwear industry in Table 3, we categorized consumer comments into different information types to further explore this aspect.

Table 3
Examples of reviews in live-streaming room.

Category of Information	Examples of Reviews	
	Chinese	English
Promotional information	优惠券怎么领?	How to get the coupon?
	有抽奖活动吗?	Is there a raffle?
	9号链接有优惠吗?	Is there a discount in link No.9?
	新会员有什么优惠?	What are the benefits for new members?
Product information	146怎么买最合算?	What is the best way to buy No.146?
	150漆皮好看还是鳄鱼皮好看?	Which looks better in link No.150, patent leather or crocodile leather?
	腿粗骑士靴主播有推荐的吗?	My legs are thick, which pair of knight boots does the streamer recommend?
	厚底乐福鞋推荐哪款?	What do you recommend for platform loafers?
	我要买长靴,推荐一下,不要高跟的。	I'd like to buy long boots. Please recommend it, I don't want high heels.
	杏色好搭衣服吗?	Does apricot color go well with clothes?

5.2. Topic model

Feature extraction. To visualize the characteristics of keywords in live comments in virtual streamers' room and human streamers' room, the keywords were extracted by adopting the commonly used TF-IDF algorithm in Python (Qaiser and Ali, 2018). Taking a brand as an example, we calculated the weight of words appearing in the comments in the situation of virtual streamers and human streamers (see Web Appendix B Figs. 1 and 2).

Clustered topics. Considering that the data set contains 30 brands, each brand contains the review data of several live broadcast rooms. We operationalized the streaming room level topic clustering in two steps. First, we clustered the similar topics between two reviews using the cosine similarity. Let θ_{bi} denote a T × R probability matrix of T topic mixtures for streaming room i of brand b with total R reviews from audience. Then, $\theta_{bi}(t, r)$stands for a topic probability of topic t for review r. It should be noted that to communicate effectively with the streamer, customer reviews in the live broadcast room are usually brief and related to only one topic. But some of the reviews covered two or more topics. After word segmentation, we used the Word2Vec model (Mikolov et al., 2013) to generate a word vector for each topic, denoted as W WTopic = (w_1, w_2 …, w_N). The semantic similarity is measured by Cosine value between two vectors. Let $\text{WTopicSimilarity}_{\text{bird}}$ denote the topic similarity between a consumer review r_1 and a consumer review r_2 for a broadcast room i of brand b on day d. We can express $\text{WTopicSimilarity}_{\text{bircd}}$ as follow:

$$\text{WTopicSimilarity}_{\text{bird}} = \frac{\sum_{t=1}^{T} \theta_{bi}(t, r_1)\theta_{bi}(t, r_2)}{\sqrt{\sum_{t=1}^{T} \theta_{bi}^2(t, r_1)}\sqrt{\sum_{t=1}^{T} \theta_{bi}^2(t, r_2)}}.$$

Second, we applied the K-means clustering techniques to the reviews processed. The K-means algorithm presented as assigning objects to the nearest cluster by distance, and the center of each resulting cluster is

Table 2
The data preprocessing process.

	Comment Type	Numbers	Examples of Chinese	Examples of English
	Total number comments in the dataset	4,061,987		
Remove: rowhead	Meaningless comments	458,248	⁂卅 。	⁂卅 。
Remove: rowhead	The repeated comments	904,988	直播间有优惠吗?(同一ID重复发送N条相同评论,N ≥ 2)	Is there any discount in the live streaming room? (The same user repeatedly sends N comments, N ≥ 2)
Remove: rowhead	Comments generated by the system	738,307	用户关注了主播。 用户分享了直播间。	Users followed the streamer. Users share the live streaming room.
	Comments for further analysis	1,960,444		

regarded as a vocabulary (Zhao et al., 2018). In this stage, the keywords were clustered into three different sets and reviews are manually selected from the results of K-means algorithm based on the following two criteria: 1) merge the semantically similar clusters; 2) extract the clusters that represented the tendency to seek information and the attitude towards the streamers of consumer. The "Word2Vec" from the "Gensim" package and the "KMeans" from "Sklearn-cluster" package in Python were used to get the results (Wang et al., 2024).

In addition, to provide a more detailed topic analysis and show the content of real-time comments sent by consumers in the live-streaming room, we employed the Latent Dirichlet Allocation (LDA) model (Blei et al., 2003) for supplementary analysis. For conciseness, the relevant contents are provided in Web Appendix B. The results from this additional LDA examination serve to corroborate the insights gained from the word2vec and k-means methods, enhancing the reliability of clustering outcomes.

5.3. Results

The Extraction and Clustering Results. Extracting and clustering the topics of consumer live comments is the first step to explore consumers' concerns about different types of information. After cleaning the comments that has nothing to do with consumers' watching live streaming and shopping behavior, we obtained three themes from the data: exchanging promotional information, exchanging product information, and interacting with streamers (not related to purchasing behavior). The extraction and clustering results were presented in Web Appendix B.

The Tendency to Seek Information. For 1537 live streaming, the type of information expressed in the comments was focused on. We calculated the proportion of promotional information comments and the proportion of product information comments in each live streaming. Therefore, we obtained 1537 records on the proportion of promotional information and 1537 records on the proportion of product information. We observed a significant effect of the streamer type on the proportion of promotional information ($M_{virtual}$ = 29.30%, SD = .19; M_{human} = 23.60%, SD = .08; t (1, 1535) = 7.49, $p < .001$). Besides, the proportion of product information was also significantly different in the two types of streamers' rooms (M_{human} = 29.03%, SD = .10; $M_{virtual}$ = 19.05%, SD = .14; t (1, 1535) = 16.07, $p < .001$). Thus, consumers pay more attention on promotional information when interacting with virtual streamers. However, Product information receives more attention in the human streamers' rooms.

5.4. Discussion

By analyzing real consumer comments data from various industries' live streaming rooms, we found initial evidence in support of H1. Consumers showed a higher tendency to seek promotional information in virtual streamers' room, while they are more concerned about product information when facing human streamers. Nonetheless, Study 1 had limitations. First, while comment text data could offer insights into consumers' intentions and attitudes as information producers (Berger et al., 2020), the brevity of consumer comments in the live streaming context limited the depth of information available for analysis. Second, we operated under the assumption that consumers' information-seeking behavior was externally driven and did not account for their inherent propensities towards seeking promotional or product information before entering the live streaming room. Third, the data in Study 1 came from a single platform (Taobao Live), and unobserved platform factors and characteristics may affect consumers' preferences over the type of information. Therefore, these improved directions motivated the next experiments.

6. Study 2

After confirming real-world validation for our proposed primary effect, Study 2 aimed to validate our theory in a controlled experimental setting. Firstly, our purpose was to demonstrate that consumers exhibit preferences for distinct types of information when interacting with different streamer types (H1). Secondly, we tested the role of motivation inference (H2). Throughout hypothesis testing, we substantiated that the impact of streamer type remains significant after controlling for consumers' inherent characteristics of seeking promotional information or product information. Finally, this study dismissed consumers' expectations of the streamer's abilities as alternative explanations. It could be argued that due to virtual streamers' perceived limitations in usefulness and dialogue (Wan and Jiang, 2023), consumers have lower expectations for virtual streamers to be able to answer complex questions and provide useful information. This may lead consumers to pose standardized questions, such as those related to promotional information. We aimed to confirm that virtual streamers heighten consumer interest in promotional information because consumers initially perceive the company adopt virtual streamers to reduce costs, rather than questioning the virtual streamers' ability to delivery product information.

6.1. Method

Participants and Design. Respondents were 430 members of Sojump (a widely used online data-collection platform) who participated for monetary remuneration (M_{age} = 30.0 years, SD = 7.96; 53.14% female). The study was designed as a single-factor (streamer type: virtual streamer vs. human streamer) between-group subjects. Participants were randomly assigned to one of two streamer scenarios. After the manipulation checks, the responses of 416 participants were used in the subsequent analysis.

Streamer Type Manipulation. Participants were instructed to engage in a skincare product shopping task within an online mall. The focal product was a moisturizing essence designed to address skin dryness during the current winter season. Participants were told that a brand conducting a live stream showcasing the moisturizing essence and they enter the live streaming room for watch. We recreated and presented visuals of the live streaming room interface, mirroring real scenarios, and included textual representations of the streamer's explanations (refer to Web Appendix C). The interface featured images of the streamer (virtual or human) as the sole distinction between the two pictures, along with links for product purchases and other relevant information.

The Tendency to Seek Information. Following participants' viewing of the live streaming interface and comprehension of the streamer's explanations, we measured their tendency to seek information by asking participants what type of information they were interested in asking the streamer. Participants were presented with six alternative questions, three pertaining to promotional information (e.g., "Is there a coupon for buying essence in this live streaming room?") and three focusing on product information (e.g., "Is this essence suitable for dry or oily skin?"). These questions were set based on common consumer inquiries observed in real live streaming scenarios. Subsequently, an open-ended question was posed after the multiple-choice questions, allowing participants to express any additional information they wanted to communicate with the streamer.

Motivation inference. Drawing on the method of previous literature (Campbell, 2007), we measured motivation inference about cost reduction using three items ("This company is reducing costs at the consumer's expense," "This company is trying to create more profits instead of serving consumers better," "This company is saving money on consumer service staff."; 1 = "disagree" and 7 = "agree"; α = .76). Motivation inference about service improvement were also measured through three items ("This company is providing better and more interesting service," "This company is trying to reduce customer's trouble in the shopping process," "This company is providing consumers with more convenient services."; 1 = "disagree" and 7 = "agree"; α = .72).

Control Variables & Alternative explanation. To verify that consumers' information-seeking tendency in live streaming rooms is exogenous, this study selected price consciousness (the degree to which consumers focus on paying low prices, see Lichtenstein et al., 1993; items such as "I am willing to go to extra effort to find lower prices"; seven-point scales, 1 = "disagree" and 7 = "agree"; α = .74; adapted from Alford and Biswas, 2002), sale proneness (the propensity to respond to a purchase offer because of the sale or discount form in which the price is presented, see Lichtenstein et al., 1993; items such as "If a product is on sale, that is a reason for me to buy it"; seven-point scales, 1 = "disagree" and 7 = "agree"; α = .82; adapted from Alford and Biswas, 2002) and product's attractiveness ("To what extent does this product attract you?"; seven-point scales, 1 = "unappealing/don't like it at all," and 7 = "appealing/like it a lot") as control variables to control the interference caused by the inherent characteristics of consumers themselves. In addition, we asked participants to report their expectation for the streamer's interactive ability ("To what extent do you expect the streamer to be able to answer your questions effectively?"; seven-point scales, 1 = "not at all" and 7 = "very expect") and the expectation for the streamer's professional ability ("To what extent do you expect the streamer to provide you with useful information?"; seven-point scales, 1 = "not at all" and 7 = "very expect") as measures of alternative interpretations. Participants completed manipulation checks for the streamer type and the product in the experimental situation. Finally, participants reported their gender and age.

6.2. Results

The Tendency to Seek Information. To calculate consumer's tendency to seek information, we created the variable by (1) totaling the number of promotional information choices selected by participants, inclusive of responses from the open-ended question, (2) totaling the number of product information selections made by participants, also including responses from the open-ended question, and (3) then subtracting the number of promotional information from the number of product information. This method of subtracting one construct from another is a valid methodology for creating a single variable that demonstrates the relative strength of one construct over another (Kim et al., 2022). Consequently, the tendency to seek information, as a variable we created, enabled us to assess the relative preferences individuals exhibit towards different types of information. Here, a positive (vs. negative) calculated value signifies that consumers demonstrate a higher tendency to seek promotional information (vs. product information).

Then, A one-way ANCOVA using price consciousness, sale proneness, product's attractiveness, the expectation for the streamer's professional ability and interactive ability as covariates revealed a main effect of the steamer type on the tendency to seek information ($F(1, 415) = 32.28$, $p < .001$; price consciousness $p = .39$, sale proneness $p = .93$, the attractiveness $p = .25$, the expectation for the streamer's professional ability $p = .88$, the expectation for the streamer's interactive ability $p = .86$). Facing with virtual streamer increased participants' tendency to seek promotional information ($M = .31$, $SD = 1.49$) more than facing with human streamer ($M = -.53$, $SD = 1.46$).

Moreover, we conducted another ANCOVA on motivation inference as a function of the streamer type, controlling for five variables mentioned above. There was a main effect of the streamer type ($M_{virtual} = .29$, $SD = 1.74$; $M_{human} = -.43$, $SD = 1.62$; $F(1, 415) = 18.18$, $p < .001$; covariate ps $> .14$). This showed that when consumers watched the virtual streamer's explanation, they tended to inference the company's motivation behind it was to reduce costs, while the motivation behind the human streamer was more likely to improve service.

Mediation. We conducted mediation analyses (Hayes, 2013; Model 4) to test whether streamer type causes consumers' inference about company's motives, which leads to a preference for different types of information. Price consciousness, sale proneness and product's attractiveness served as covariates in the mediation analysis models to ensure that any mediation effects we observed are not disrupted. Bootstrapping confirmed the indirect effect of streamer type on the tendency to seek information (10,000 draws; indirect effect = .46, SE = .11, CI_{95} = [.2505, .6913]), in support of our theoretical model. Besides, to test whether the expectation of the streamer's professional and interactive abilities might explain the impact of streamer type on information preference, we included these two variables as mediator respectively and repeated the above analysis. The results substantiated our assumption that both the expectation of professional ability (indirect effect = .0006, SE = .01, CI_{95} = [-.0145, .0208]) and the expectation of interactive ability (indirect effect = −.01, SE = .02, CI_{95} = [-.0446, .0319]) did not mediate the streamer type effect on information preference.

6.3. Discussion

Study 2 replicated the impact of streamer type on consumers' information-seeking behavior within a controlled setting and offered mediation support of our model. Encountering a virtual streamer leads consumers to infer that the company prioritizes cost reduction, consequently heightening consumers' tendency to seek promotional information. Moreover, consumers' preferences for different types of information were not mediated by expectations about streamers' abilities. Subsequent studies will use specific product categories to illustrate how both streamer types leverage their respective strengths and align with consumers' shopping psychology to drive sales growth.

7. Study 3

Study 3 aimed to elucidate the type of products that virtual streamers are adept at presenting to enhance company sales. The goals of this study were threefold. Firstly, building upon the findings of Studies 1 and 2, which affirmed that virtual streamers excel in presenting promotional information aligned with consumer preferences, we tested the efficacy of virtual streamers in driving sales through promotional product. Secondly, we ascertained the generalizability of the effect by using a different product type: a backpack. Thirdly, to eliminate the possible interference of the streamer's gender on the results, both the virtual and human streamers in this study were portrayed as male characters.

7.1. Method

Participants and Design. 620 respondents ($M_{age} = 30.70$ years, $SD = 8.39$; 57.91% female) from Sojump participated in exchange for monetary compensation in a 2 (streamer type: virtual streamer vs. human streamer) × 2 (product category: promotional product vs. non-promotional product) between-subjects design. 7 respondents failed the manipulation checks, so the responses from 613 participants were used for subsequent analysis.

Streamer Type Manipulation. Participants were tasked with purchasing a backpack from an online mall. They were presented with an image of the backpack and informed about a brand conducting a live streaming for the product. We manipulated the streamer type, either human or virtual, by presenting two different live streaming room images. As in Study 2, the only distinction between the two images was the presence of a human or virtual streamer. All other components (the product appearance and information, the link to buy the product, comments from other consumers, etc.) remained consistent between the two versions. We also provided the content of the streamer's explanation (see Web Appendix C).

Product Category Manipulation. We manipulated promotional product versus non-promotional product by providing obvious promotional labels to participants. In the promotional product condition, participants were presented with a prominent tag indicating: "[Exclusive offer in the live streaming room] Place your order through the exclusive product link in the live streaming room and get 15% off." And

for the non-promotional product condition, participants did not encounter any promotional advertisements.

Measures of dependent, mediating and control variables. Following the above steps, we informed participants that ten people would receive a CNY300 bonus payment and asked whether they were willing to use their all or part of bonus to buy the backpack. The amount paid by the participants (a sliding scale ranging from CNY0 to CNY300) served as our dependent variable. Then, participants answered a set of questions about their tendency to seek information, which were nearly identical to those used in Study 2, except for the details of three promotional information choices and three product information choices, which were tailored for Study 3. Specifically, we adjusted questions about product information for this backpack (e.g. "Can this backpack hold a computer?") and questions about promotional information options (e.g. "Are there any freebies with the purchase?"). Notably, questions related to specific discount percentages or amounts were not included in the promotional information options, as Study 3 offered participants a 15% discount on the item in the promotional product condition. Next, participants answered three items related to motivation inference about cost reduction ($\alpha = .77$) and three items about service improvement ($\alpha = .86$), which were consistent with previous study.

Moreover, for control purposes, we measured participants' price consciousness ($\alpha = .85$), sale proneness ($\alpha = .71$), the expectation for the streamer's interactive ability, the expectation for the streamer's professional ability, and product's attractiveness by using the same scales as in Study 2. At the end of the survey, we conducted manipulation checks for the streamer type and product category and collected demographic information (gender and age).

7.2. Results

The Tendency to Seek Information. We ran an analysis of ANCOVA entering streamer type (1 = virtual streamer; 0 = human streamer) as independent variable, and five variables as covariates (price consciousness, sale proneness, the expectation for the streamer's interactive ability, the expectation for the streamer's professional ability, product's attractiveness) as covariates predicting the tendency to seek information. The results showed a significant effect of streamer type ($M_{virtual} = .32$, $SE = 1.73$; $M_{human} = -.55$, $SE = 1.36$; $F(1, 612) = 45.95$, $p < .001$; price consciousness $p = .36$, sale proneness $p = .18$, the expectation for the interactive ability $p = .71$, the expectation for the professional ability $p = .65$, the attractiveness $p = .99$). For the sake of brevity, the analysis of the effect of streamer type on motivation inference and the analysis of mediating effect were not discussed here. Web Appendix C contains these supplementary analyses.

Effects of streamer type and product category on purchase behavior. Next, to test the moderating effect of product category, we used a regression analysis with purchase behavior as the dependent variable and streamer type, product category (1 = promotional product, 0 = non-promotion product), and their interaction as independent variables. Neither the main effect of streamer type ($b = -2.19$, $SE = 3.18$, $t(609) = -.69$, $p = .50$) nor product category ($b = -1.92$, $SE = 3.16$, $t(609) = -.61$ $p = .54$) was significant. Most importantly, in support of H3, this main effect was qualified by a significant interaction in the expected direction ($b = 9.90$, $SE = 4.48$, $t(609) = 2.21$, $p = .03$). To illustrate the role of virtual streamers in recommending promotional products was more prominent than that of human streamers, we examined the effect of discounts in the virtual streamer and human streamer condition with planned contrasts (see Fig. 3). In the virtual condition, the participants were willing to pay more for the promotional product ($M = 66.50$, $SD = 31.25$) compared to the non-promotional product ($M = 58.51$, $SD = 26.19$, $t(303) = 2.42$, $p = .02$). However, when human streamer introduce product to consumer, the amount paid by participants was similar regardless of whether the product was being promoted ($M = 58.78$, $SD = 26.83$) or not ($M = 60.70$, $SD = 26.15$, $t(306) = .64$, $p = .53$).

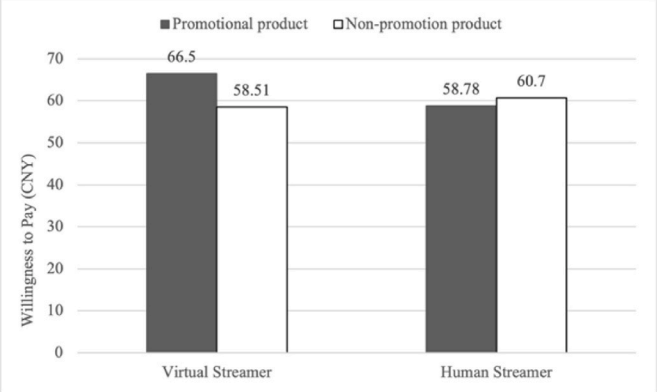


Fig. 3. Effects of streamer type and product category on willingness to pay in Study 3.

7.3. Discussion

The results of Study 3 revealed that since consumers show a higher tendency to seek promotional information when presented with explanations from virtual streamers, they increased purchase intent when virtual streamers recommend products featuring notable promotional content (e.g., promotional items). In the subsequent study, we examined which product type showcased by the human streamer are more effective in driving sales growth.

8. Study 4

Having substantiated the impact of human streamers on improving service intuition, Study 4 examined how human streamers introduce products to consumers effectively. We hypothesized that human streamers recommend products will lead to greater consumer interest and expectations about the inherent characteristics of the product (e.g., quality, novelty). Since consumers have a greater demand for information about the inherent characteristics of new products (vs. non-new products), we anticipated that human streamers introducing new products would be more successful in boosting sales. In addition, Study 4 adopted the different measures of motivation inference and purchase intention from previous experiments to enhance the robustness of the research.

8.1. Method

Participants and Design. 620 respondents ($M_{age} = 33.25$ years, $SD = 6.60$; 60.5% female) from Sojump participated in a 2 (streamer type: virtual streamer vs. human streamer) × 2 (new product vs. non-new product) between-subjects experiment. They were presented with a fictitious brand of smart household appliances and was told that the brand was introducing floor sweeping robots in the live streaming room. After excluding samples that did not pass the manipulation checks, the responses of 600 participants were used for further analysis.

Manipulations. We selected a sweeper robot as focal product. Participants were informed that they were considering purchasing a sweeper robot for daily home cleaning requirements and a brand was conducting a live streaming about the robot. Subsequently, participants were shown the picture representing the live streaming room, with the image of a virtual streamer or a human streamer, following the format of Study 2 and Study 3. Besides, Participants assigned to the new product category were explicitly informed that the sweeper robot was a novel addition to the market through an obvious label (see Web Appendix C).

Measures of dependent, mediating and control variables. Participants reported their purchase intention for the sweeping robot on a three-item scale ("I will buy the product that this streamer promotes in

the live streaming," "I intend to purchase the products that this streamer promotes in the live streaming," "I will consider this streamer's live streaming room as my first shopping choice"; 1 = "disagree" and 7 = "agree"; α = .90; adapted from Gao et al., 2023). Then, we measured the tendency to seek information according to the method used in previous studies. Next, draw on prior research (Burman et al., 2024), an open-ended question was used to measure participants' motivational inferences about the company. Participants were asked to write down what they thought was the reason for the company to schedule this streamer to introduce the product. The written responses were coded by two independent raters (1 = improve service, −1 = reduce costs, 0 = improve service and reduce costs; Cohen's κ = 1). Finally, participants responded to questions including control variables (the expectation for the streamer's interactive ability, the expectation for the streamer's professional ability, price consciousness (α = .87), sale proneness (α = .76), the product's attractiveness). Participants also completed the streamer type check, product category check and indicated their gender and age.

8.2. Results

Main Effects. We first examined the psychological activities of consumers when they watch different types of streamers (1 = human streamer; 0 = virtual streamer) introducing the floor sweeping robot. A one-way ANOVA on the tendency to seek information, controlling for five variables (price consciousness, sale proneness, the expectation for the streamer's interactive ability, the expectation for the streamer's professional ability, product's attractiveness), yielded the expected main effect of streamer type (M_{human} = .51, SD = 1.45; $M_{virtual}$ = −.29, SD = 1.72; F (1, 599) = 36.44, p < .001; covariate ps > .16). This indicated that participants facing with human streamer showed the higher tendency to seek product information than those facing with virtual streamer. Web Appendix C contains additional analyses of motivation inference and mediating effects.

Effects of streamer type and product category on purchase intent. We regressed purchase intent on streamer type, product category (1 = new product, 0 = non-new product), and their interaction. The main effect of streamer type was not significant (b = −.06, SE = .12, t (596) = −.45, p = .65), as was the product category (b = .05, SE = .12, t (596) = .43, p = .67). Importantly, the effect bought by the interaction is significant (b = .40, SE = .18, t (596) = 2.29, p = .02), which supports H4. Subsequently, we examined the consumers' purchase intentions when the human and virtual streamers introduce different product categories separately (see Fig. 4). In the human condition, consumers had a higher purchase intention when streamer explained new product (M_{new} = 5.37, SD = .81; $M_{non\text{-}new}$ = 4.91, SD = 1.06; t (297) = 4.19, p < .001). However, in virtual condition, there was no significant difference in purchase intention whether it was a new product or not (M_{new} = 5.02, SD = 1.11; $M_{non\text{-}new}$ = 4.97, SD = 1.28; t (299) = .38, p = .70).



Fig. 4. Effects of streamer type and product category on purchase intent in Study 4.

8.3. Discussion

Study 4 not only provided evidence that consumers pay more attention to product information when interacting with human streamers, but also confirmed that the introduction of products with significant product details (particularly new products) by human streamers positively influences consumers' purchase intent.

9. Study 5

This study consists of qualitative consumer interviews, designed to gain a deeper understanding of consumers' perception and psychological reactions to virtual and human streamers. Based on this, Study 5 had two objectives. First, the interview data further confirms our findings from the four previous studies and combines them with consumer experiences (Agnihotri and Bhattacharya, 2024). Second, consumer perspectives can provide us with rich additional insights about two types of streamers. Therefore, we recruited 20 participants with experience interacting with virtual and human streamers and conducted half-structured interviews with them.

9.1. Method

We used a purposeful sampling approach and invited respondents from online platforms to take part in interviews about live streaming shopping and interaction with streamers. Each interview lasted an average of 31.5 min and the respondents included 11 females.

All interviews followed a half-structured interview guide (Arsel, 2017). During the interviews, participants described four primary aspects: a) Recounting a recent live streaming shopping experience, detailing the items purchased, the reasons behind their choices, and the streamer's influence in the process; b) Recalling their initial encounters with human and virtual streamers, as well as experiences when interacting with both types of streamers on a daily basis; c) Assessing human and virtual streamers and narrating their shopping experiences in their rooms; and d) Offering insights on the future development of virtual and human streamers. The interviews were transcribed using iflyrec software and independently checked by two researchers. Further details regarding the interview guide and participant demographics are available in Web Appendix D.

9.2. Results

All respondents indicated that the information provided by streamers played a key role in shopping decisions. On the one hand, human streamers promote purchases by reducing uncertainty through the provision of product information. As respondent 1 (R1, similarly hereinafter) stated, "Compared to the traditional form of online shopping, human streamers can give me vivid feedback on product information and reduce the risk that the product may not be suitable for me." Some respondents noted that the product information presented by human streamers facilitated decision-making for consumers, as R4 mentioned, "Human streamers streamlined my product selection process. I don't need to repeatedly compare the information on the web but listen to the streamer to compare the advantages and disadvantages of similar products to make a shopping choice. This makes my shopping more efficient." On the other hand, virtual streamers help consumers place orders quickly by providing clear and precise promotional information. R9 said, "Firms like to play 'word games' with promotions in order to attract consumers' attention. Human streamers even complicate the game because they introduce all kinds of promotions and may bundle other products. So, I prefer virtual streamers, who have the potential to summarize information, and can provide promotional information

quickly and clearly without increasing my cognitive costs."

Respondents shared their initial reactions upon encountering virtual streamers. They described feelings of surprise and interesting towards virtual streamers, with 90% of respondents indicating their first impression was that the company was working to reduce costs. For instance, R13 said, "Upon my first interaction with virtual streamers, I was surprised. It piqued my curiosity as it deviated from the norm, prompting me to consider the meaning and reasons behind integrating AI into the live-streaming environment." Similarly, R6 remarked, "I thought it was quite interesting to meet the virtual streamer. I even engaged in light-hearted banter, requesting a song performance." Nevertheless, some respondents expressed confusion, as exemplified by a respondent who stated, "Initially, I was perplexed when introduced to the virtual streamer. The company's human streamers had proven effective, leading me to question the necessity of employing AI. Subsequently, I rationalized that the industry trend is just like this, and there is a need to save costs, notably labor-related expenses."

When discussing their interactions with streamers, respondents noted that virtual streamers' ability to provide prompt and precise responses enhanced the consumer experience in acquiring information, and they paid more attention to promotional information during interactions. For example, R2 remarked, "A human streamer repeats the advantages and disadvantages of the products over and over again, and there is no way to accurately select my questions and give answers among a group of consumers. In contrast, the virtual streamer's algorithm-driven swift responses are something that human don't have." R5 shared, "When virtual streamers emphasize product uniqueness and excellence, I don't really care about that information. I just want to know if the product is competitively priced. After all, the role itself is meant to reduce costs and increase efficiency." As for human streamers, respondents said they are more helpful to consumers in choosing the right product and respondents showing greater interest in product information during interactions. As expressed by R18, "Previously, my understanding of collagen proteins was limited, but a human streamer elucidated the differences effectively and aided in selecting the most suitable product, reflecting the company's commitment to service quality." Echoing this sentiment, R14 summarized her interaction preference, stating, "I prioritize product experience and quality when engaging with human streamers, and after then I may consider the price." These insights are consistent with our H1.

More interestingly, respondents expressed the view that when a live streamer highlights promotional information, it could create a negative perception of the product. In this regard, R9 explains, "Given the higher costs associated with human streamers compared to virtual ones, it raises questions on why it is possible to run promotions despite the increased costs and still keep the company profitable. This suggests that the actual product cost may be lower than perceived, potentially indicating reduced quality. I don't want to spend my money on a low-quality product." This observation aligns with our hypotheses that consumers associate virtual streamers with cost reduction, justifying promotional content presentation. However, human streamers increase the company's labor costs, and at this point, they should focus on meeting consumers' expectations regarding product quality and providing comprehensive product information, rather than emphasizing promotional activities.

In streamer evaluations, all respondents perceived the utilization of virtual streamers as a cost-reduction motivation, with 45% expressing negative sentiments. R11 remarked, "The use of virtual streamers is a very frugal and perfunctory way of live streaming." Similarly, R15 noted, "Virtual streamers serve a strictly business purpose." Offering a different perspective, R3 stated, "Virtual streamers essentially act as company-selected spokespersons. My assessment of virtual streamers extends to the company's stance. When a virtual streamer is adopted to speak about a product, I don't think the company is willing to pay a higher cost for the consumer." In addition to these views, 55% of respondents reported positive attitudes. Some respondents said, "Virtual streamers symbolize the company's proactive embrace of new technology. While the company is trying to control costs, this move is not hard to accept." "I prefer interacting with virtual streamers. It benefits both the company and me in terms of cost savings and time efficiency. It can actually be a win-win situation."

When evaluating human streamers, respondents generally praised the detailed and vivid product information provided, attributing this to an enhanced service experience. As stated by R10, "Virtual streamers, while capable of product demonstrations, lack the realism and flexibility of human streamers. A real person can better understand my concerns about a product not suitable for me and provide targeted explanations, fostering a sense of trust." R12 said, "Sometimes I have questions about a product, and that's when I need a human streamer to perform the service. Because having a question means I have uncertainty about this product, and that uncertainty can only be resolved by a human. I don't think an AI can empathize with me." R7 stated, "Human streamers surpass static online presentations by vividly showcasing products and conveying extensive information, creating a more authentic experience for consumers." These evaluations confirm H2 of this research.

Respondents highlighted distinctions in the purchasing experience between virtual streamer and human streamer environments. 90% of respondents expected virtual streamers to recommend more cost-effective products. For instance, R8 noted, "Virtual streamers are helping you return your goal to the shopping task itself and buy a more cost-effective product. This relates to the attributes of AI, where you naturally think that way." Similarly, R16 mentioned, "The absence of human streamers suggests a reduced emphasis on product presentation costs. Since the cost of the product has gone down, its price should also be relatively lower or have some big discounts." This underscores the belief that regardless of consumers' positive or negative perceptions of virtual streamers, virtual streamers should give full consideration to recommending promotional products. This confirms H3. In addition, respondents cited diverse reasons for purchasing products in human streamer's room, including access to comprehensive product information, the streamer's expertise enhancing product appeal, and persuasive language prompting impulse purchase. R7 stated, "When the streamer has finished the introduction and I am sure that this product is suitable for me, I make immediate purchase decisions." R20 mentioned, "There may be many reasons to shop in a human streamer's room, such as buying for the personality and humor of the streamer or buying because I learned more." R17 expresses feelings related to impulse buying, "Human streamers often say things that stimulate consumers to purchase, such as constantly talking about how great this product is, led me to believe that I had a need for it. But after making the purchase, I realized I didn't actually need it."

Several respondents opined, "As technology advances, virtual streamers will be more advantageous in introducing classic products. With vast consumer data on classic products, AI can analyze and objectively present this information to consumers, aiding purchase decisions significantly. Conversely, new products lacking consumer data may be better promoted by human streamers, given the fundamental difference between virtual and human streamers. The unverified information said by human streamers will be more credible." This viewpoint is consistent with H4 in our research. Some respondents evaluated this scenario from a cost-benefit standpoint, stating, "In the future, a symbiotic relationship between virtual and human streamers will be essential. Human streamers entail higher costs, particularly burdensome for smaller companies employing multiple human streamers. Therefore, the cost-effective approach lies in utilizing virtual streamers alongside human counterparts." Furthermore, concerns were raised regarding the advancement of AI technology and virtual streamers. R3 said, "I will feel a bit scared if virtual streamers really do not differ from real people in terms of their appearance, interaction capabilities, and so on. This could evoke a new form of the Uncanny Valley effect."

9.3. Discussion

The analysis of the interviews confirmed all the findings of Studies 1–4 and improved the ecological validity of the experiment. Consumers' experience of interacting with the streamers supports H1 that they pay more attention to promotional information (vs. product information) when facing with virtual streamers (vs. human streamers). This is attributed to consumers' perception about the company's motivation, as suggested by respondents' evaluations of the streamers (supporting H2). It is clear from the shopping experience in the different streamers' room as emphasized by respondents that they expect virtual streamers to introduce promotional products (supporting H3). And human streamers would be more advantageous to introduce new products (supporting H4).

10. General discussion

10.1. Conclusions

In the realm of LSC, the introduction of virtual streamers enables consumers to engage in live streaming to explore products and engage socially at any time, offering a convenient, seamless, and customer-friendly experience while reducing labor and financial costs for companies. Previous research has highlighted potential negative implications of AI agents in various contexts (e.g. Sands et al., 2022; Luo et al., 2019; Stein and Ohler, 2017). However, this research focuses on the impact of virtual streamers with digital avatars on consumers' shopping mindset within LSC, linking this impact to a positive increase in product sales.

Using a dataset of consumer comments in real live streaming rooms, three experiments and half-structured interviews, we find that virtual streamers and human streamers evoke distinct shopping goals in consumers. Besides, the role of virtual streamers in driving product sales should not be ignored by companies. Consumers who encounter virtual streamers (vs. human streamers) actively show the preference for promotional information (vs. product information), indicating a focus on "saving money" (vs. product suitability) during their shopping journey. This shift in consumer mindset is driven by consumers' inferences about the company's motives, with consumers more likely to believe that the use of virtual streamers aims to reduce costs rather than improve service. Consequently, when virtual streamers present products with prominent promotional content, it effectively boosts product sales.

10.2. Theoretical contributions

Our research contributes to a better theoretical understanding of consumer responses to virtual streamers. The first theoretical contribution concerns the literature on the psychology of human-AI interaction (Puntoni et al., 2021; Davenport et al., 2020). We extend this type of research by demonstrating for the first time that consumers' interactions with virtual streamers led to a higher tendency to seek promotional information. While existing research has addressed various psychological responses of consumers towards AI agents, including trust (e.g., Mostafa and Kasamani, 2022), perceived empathy (e.g., Liu-Thompkins et al., 2022; Luo et al., 2019), algorithmic aversion (e.g., Stein and Ohler, 2017), and emotional reactions (e.g., Tran et al., 2021), there has been a lack of focus on consumer shopping psychology during interactions with AI. In addition, virtual streamers, as a prominent category of AI agents, have garnered scholarly interest. Recent research has started to investigate the impact of virtual streamers on consumer trust (Chen et al., 2024), perceived capabilities (Yan et al., 2024; Sun and Tang, 2024; Wan and Jiang, 2023), perceived social presence (Gao et al., 2023; Wang et al., 2023; Wang et al., 2023), and other psychological responses. However, to the best of our knowledge, consumer expectations for information provided by virtual streamers have not been explored. Therefore, the current research bridges these gaps by examining consumer's "save money" mindset when engaging with virtual streamers, contrasting with their focus on product information when interacting with human streamers. This distinction enables us to capture the differentiated information-seeking behaviors exhibited by consumers in distinct live streaming rooms, offering insights into consumers' shopping mindset when dealing with AI agents and their perceptions of the role of virtual streamers.

We present a rich theoretical framework to explain consumers' information-seeking behavior in live streaming rooms. Specifically, consumers interpret companies' utilization of virtual streamers as prioritizing cost reduction, promoting them to seek compensation for perceived financial benefits. This finding offers a deeper perspective on understanding consumers' psychological responses to virtual streamers. More importantly, by analyzing this underlying psychological mechanism, we extend previous research on consumers' attitudes towards companies' incorporation of AI technologies. Prior research has underscored that consumers associate company initiatives to adopt AI technologies with cost reduction (Castelo et al., 2023; Sands et al., 2022; Ling et al., 2021; Williams, 2019). Our findings are consistent with this viewpoint given that virtual streamers fall under the purview of AI agents. Through an exploration of consumer shopping psychology and its underlying mechanisms, we further illustrate a key consequence of consumer's motivation inference: consumers' perception that companies intend to reduce costs through AI technology triggers perceptions of price unfairness and prompts a pursuit of financial compensation. This aspect remains underexplored in prior literature. Thus, by unveiling these underlying psychological dynamics, we elucidate how consumer perceptions of AI technology influence their shopping behaviors in live streaming environments.

The second theoretical contribution relates to the literature on AI recommends products (Longoni and Cian, 2022). We delve into the conditions that enhance the effectiveness of AI recommendations and communication with consumers, particularly when virtual streamers emphasize products with significant promotional information. While past research has primarily examined AI's efficacy in recommending utilitarian (e.g., Yan et al., 2024; Longoni and Cian, 2022) or search products (e.g., Xie et al., 2022), our framework is anchored in consumers' information preferences in diverse live-streaming rooms. Through a blend of quantitative and qualitative methods, we elucidate that virtual streamers are better suited to recommend promotional products, while human streamers are adept at introducing new items. Our findings provide a new perspective for AI's product recommendation strategy.

By aligning virtual streamers with consumer expectations for promotional content, companies can potentially alleviate resistance to AI recommendations. Prior research has extensively studied consumer resistance to AI recommendations due to the prevalence of "algorithm aversion" (e.g., Crolic et al., 2022; Luo et al., 2019). Our findings suggest that virtual streamers recommending promotional products can make consumers feel that they benefit from interacting with AI, potentially fostering greater acceptance of AI recommendations.

Finally, we also contribute to the literature on LSC and virtual streamers (Gao et al., 2023; Lu and Chen, 2021). The literature in this area has examined factors influencing the marketing effectiveness of human streamers (e.g., streamer characteristics, interactivity, trust; Zhang et al., 2022; Guo et al., 2022; Lu and Chen, 2021), limited attention has been given to the impact of virtual streamers on the consumer experience. We enrich the discussion on virtual streamers and expand this rising field by examining consumer responses to them. Moreover, existing research on virtual streamers has explored the role of factors such as streamer appearance, language, and emotional expression (e.g., Chen et al., 2024; Yao et al., 2024; Xie et al., 2024; Sun and Tang, 2024; Hu and Ma, 2023). These literatures have examined how the obvious characteristics of virtual streamers influence consumers' purchase intention or willingness to accept. However, our focus is on elucidating the impact of consumers' inherent subjective impression of

virtual streamers, notably their perception of cost-effectiveness, on information preferences. This perspective adds depth to discussions on consumer perceptions of virtual streamers. Finally, it is worth noting that there remains a gap in the existing literature in understanding of how the difference between virtual streamers and human identities affects consumer purchases. By conducting a comparative analysis of the two streamer types, our research offers empirical evidence to elucidate the advantages of streamers in driving sales.

10.3. Managerial implications

The widespread deployment of AI in online environments offers new ways for companies to optimize customer service, and virtual streamers directly convey brand dynamics to consumers. Therefore, companies utilizing or considering the use of virtual streamers should grasp consumer perceptions and interactions with these agents. In this context, our findings carried direct managerial implications, proposing a set of guidelines for the effective deployment of virtual streamers.

Our findings provided useful insights into how companies can optimize their product presentation strategies through streamers. Rather than regarding virtual streamers merely as simple substitutes for human streamers, companies should carefully monitor consumer live comments and their perceptions of both types of streamers. Given that virtual and human streamers evoke different subjective responses from consumers, they should be distinguished and used in tandem for delivering information effectively. Virtual streamers can be more effective in promoting items with prominent promotional information but are not suitable for recommending new products.

This research indicates that virtual streamers have great potential to impact key marketing outcomes in LSC. We find that companies can enhance consumer satisfaction by having virtual streamers focus on products with significant promotional information and increasing interaction frequency during promotional periods. Such moves help to match consumers' expectations for promotional information. Additionally, compared with virtual streamers, human streamers should play a central role in explaining new products to capitalize on consumer interest and attention.

According to our findings, consumers' higher tendency to seek promotional information when exposed to virtual streamers is caused by their motivation inferences about the companies. This activates consumers' cost-conscious shopping mindset and the pursuit of monetary compensation. To address this, another practical implication is that marketers should carefully organize the linguistic framework of virtual streamers to facilitate effective interactions. It is essential to attach important to the language and information transmission of virtual streamers, otherwise consumers may develop other negative perceptions, such as questioning the usefulness of virtual streamers (Wan and Jiang, 2023) or believing that they are not taken seriously by the company. In response to this dilemma, companies can adjust the initial greeting of virtual streamer, explicitly conveying their role as helpful assistants focused on facilitating cost-effective purchases and decision-making. For instance, a virtual streamer can say hello like this, "I am an assistant who answers your questions in a timely manner. I am better suited to help you buy a cost-effective product or make the most economical decision." This type of greeting can inform consumers that they are interacting with a company' carefully placed and targeted virtual streamer, thus helping to mitigate potential negative perceptions and provide clarity on shopping goals.

Our exploration of consumer shopping psychology within live streaming environments also offers valuable insights for e-commerce platforms in devising effective streamer selection strategies. Leading platforms like Amazon, Taobao, Pinduoduo, and Temu are actively embracing LSC. Taking Pinduoduo and Temu as exemplars, these platforms prioritize offering consumers competitive pricing and discounted items. Our research indicates that virtual streamers align well with consumer expectations when promoting promotional products. Therefore, businesses operating on these platforms may benefit from leveraging virtual streamers to meet consumer demands effectively, leading to cost savings and enhanced operational efficiency.

10.4. Limitations and future research

This research has some limitations that may provide opportunities for future research. First, our empirical context is Taobao live in China. But LSC and virtual streamer on other platforms such as TikTok and Facebook also attract a large number of consumers. As live streaming development strategies vary across platforms, this may shape different styles of virtual streamers. Future research may consider the performance of virtual streamers on other platforms and explore on product recommendation strategies and differences with live streamers. Second, due to Taobao Live's protection of user privacy, we did not observe consumer characteristics (e.g., age, gender) in our sample. However, taking consumer profiling into account helps to harvest insights on how consumer characteristics affect the effectiveness of virtual or human streamers' recommendations. Market research on live streaming can provide an initial perspective on this point. The survey of consumers who watch virtual streamers (N = 1654) in 2023 showed that the users were mainly consumers living in first-tier cities (23.8%), new first-tier cities (32.5%) and second-tier cities (24.9%). The majority of consumers were between 20 and 40 years old and more female (61.5%) (iiMedia Research, 2023). The reason for this may be that the communication form of virtual streamers is novel and youthful, attracting a large number of urban consumers (iiMedia Research, 2023). Future research can collect relevant data to research the influence of consumer characteristics.

In the real data of Study 1, we removed system-generated comments during preprocessing. While these comments reflect various consumer behaviors like following and sharing live-streaming rooms, they are not directly relevant to our current research focus. However, exploring how consumer interactions with virtual or human streamers impact loyalty and sharing behavior is crucial for future investigations. It is also necessary to understand how a particular performance of the streamer at a given moment can increase consumer interest in room. Here, consumer's following and sharing behavior may be appropriate proxy variables, and these data can be obtained from live comments. Therefore, it would be interesting to enhance the observation of consumer behavior in the live-streaming room and match it with the dynamics of the streamer.

The development of virtual streamers typifies the growing sophistication of AI technology, with virtual entities evolving toward a high degree of similarity to humans. For example, Silicon Intelligence has developed and implemented AI digital humans that can be used in live streaming scenarios, which realistically reproducing the form, movements, voice of a real person, and behaving as if they were a real person in a live streaming room (Silicon Intelligence, 2023). When digital bodies are sufficient to deepen their understanding and interactions with consumers through deep learning programs, consumer's service evaluations may change and clearly perceive the company's efforts to implement high-quality AI agents. Meanwhile, as AI approaches the limits of its similarity to humans, advanced digital bodies may have the effect of "the uncanny valley of mind" (Stein and Ohler, 2017). It will be interesting to discuss these possible changes in the future, and companies also need to weigh the application of creative technology against possible outcomes.

Looking ahead, virtual streamers are likely to gain increased acceptance as AI capabilities in simulation and performance improve. Exploring contexts where consumers are more receptive or resistant to virtual streamers will be valuable. AI has the potential to transform sales processes across various stages, from finding prospects to pre-approaching, which leads to making presentations to follow-up (Syam and Sharma, 2018). This will require marketers to adapt and innovate in understanding and predicting consumer behavior through AI. At least in

the realm of LSC, significant potential for enhancing virtual streaming optimization remains to be exploited.

To sum up, virtual streamers represent a novel application of AI-based avatars in live streaming environments, offering companies benefits such as enhanced efficiency, cost reduction, and improved consumer interactions. Our study shed light on the evolving dynamics of the interaction between consumers and companies in live streaming environments and proposed a specific consumer mindset when encountering virtual streamers. By analyzing customer comments and activities in live streaming rooms, companies can devise strategies to enhance sales performance through AI technologies.

CRediT authorship contribution statement

Yaping Chang: Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Han Wang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhenjiang Guo:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Formal analysis, Data curation.

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Declaration of competing interest

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Appendix A. Supplementary data

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Data availability

Data will be made available on request.

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